Matt *Kerkstra*

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SUMMARY

Staff-level AI engineer with seven years building customer-facing production systems. At VideaHealth, I operate at Staff scope across 45 engineers, owning LLM product delivery from evaluation and retrieval through KServe/vLLM serving, Kubernetes operations, backend integrations, safe rollout, and user feedback. Cut clinician edit rate from roughly 60% to 40%, shipped production AI in 16 days, and enable six engineers across 13 namespaces and nine services. TypeScript, Python, Go, React, PostgreSQL.

CORE SKILLS

ML / LLM	Kubeflow, KServe, vLLM, LangGraph, Langfuse, RAG, Milvus, BAAI/bge-m3, Presidio, AWS Bedrock, Anthropic Claude, MCP
Infrastructure	Kubernetes, Istio (ambient mesh), ArgoCD, Helm, Kustomize, Terraform, Karpenter, GitHub Actions, cert-manager, external-secrets
Data	PostgreSQL, PostGIS, Aurora, ClickHouse, Snowflake, Drizzle ORM, Kysely, SQLAlchemy, Mongo -> PG migrations, Redis / Valkey, BullMQ, Kafka
Languages	TypeScript, Python, Go, SQL
App Stack	NestJS, Fastify, FastAPI, Pydantic, React, Next.js, Jotai, React Query, OPFS, Service Workers
Observability	Grafana, Tempo, Alloy, OpenTelemetry, Mimir, Loki, Pyroscope

EXPERIENCE

Senior Software Engineer *at* **VideaHealth** · Remote

May 2023 - Present

Operating at Staff scope across a 45-person engineering org: own production LLM products, the Kubernetes ML platform, and source-system integrations.

- Co-led AI product delivery for **Canon**, a block-based LLM system that learns each customer's structure. Evidence-gated evaluation and rollout - deterministic diffs, LLM-as-judge, clinician calibration - cut edit rate from roughly 60% to 40%.
- Shipped the first production **LLM system** with two teammates in 16 days across 3 services: checkpointed graph, embeddings + Milvus retrieval, PHI anonymization, SSE streaming, and review UI.
- Built the **AI evaluation and feedback loop**: edits attributed to source blocks, per-block performance stats, and pattern dashboards that let measured user behavior prioritize improvements.
- Own the **Kubernetes ML platform**: Istio ambient mesh, ArgoCD, KServe + vLLM model serving, Langfuse, and Milvus across 13 namespaces / 9 services. Cost stayed flat; six engineers contribute regularly.
- Designed a **production shadow evaluation** on sampled sessions: both variants run, delivered notes never wait, and activation remains reviewer-gated.
- Led the **MongoDB -> PostgreSQL/PostGIS** migration of 100M+ records; cut heavy queries from >5s to <500ms. Own schemas, Python algorithms, and TypeScript integrations across 10+ versions.
- Built the end-to-end **voice-to-clinical-note product**: offline-resilient React capture, BullMQ async processing, transcription, LLM summarization, and PMS context.
- Built the unified-appointments backend behind the **largest U.S. DSO contract**, including vault search, dual-mode practices, and legacy PMS optimizations.
- Run the bi-weekly **Backend Guild**: org-wide typed-query patterns, AI developer tooling, and written ADRs.

Software Engineer *at* **Paperspace** · Remote

Jun 2022 - Mar 2023

- Rebuilt Node.js + Stripe billing to support **2.5× YoY revenue growth** while keeping payments under 200 ms.
- Implemented real-time fraud and sanctions checks that shut down illicit GPU crypto-mining from embargoed regions, **reducing chargebacks >50%**.

Senior Software Engineer / Software Engineer *at* **Hotel Engine** · Remote

Jul 2021 - Jun 2022

- Introduced bundle splitting, CDN routing, and feature flags, shrinking **mean deploy time from 15 min to 6 min** for 40+ engineers.
- Drove Redux -> React Query migration, reducing cold-start data fetches 40% and bundle size 20%.

Software Developer *at* **Reynolds & Reynolds** · Houston, TX

Feb 2019 - Jun 2021

- Converted a 20-year-old version-control system for F&I forms from VB6/SQL to COBOL + Pick BASIC - responsible for distributed delivery, usage tracking & billing, and integration with F&I systems. Assumed lead role two weeks after onboarding and delivered on schedule.

SIDE PROJECTS

Narrative Nexus *at* **narrative.sh**

TypeScript · Next.js · Go · Postgres/pgvector · Yjs

AI-powered companion for tabletop RPG game masters. Co-built with a group of long-time friends who are also software devs.

EDUCATION

Rice University · B.A.

2018 · GPA 3.71